Revision Notes for Class 8 Science Chapter 10 – Light: Mirrors and Lenses

Light helps us see the world around us by reflecting off surfaces and entering our eyes. In this chapter, you explore how mirrors and lenses change the path of light to form different kinds of images. Everyday observations, such as distorted faces in funhouse mirrors or warnings on vehicle mirrors, bring out the importance of understanding light, mirrors, and lenses.

Spherical Mirrors: Concave and Convex

Curved mirrors, known as spherical mirrors, are created by shaping glass into part of a sphere. There are two types: **concave mirrors** (curved inwards, like the inner surface of a spoon) and **convex mirrors** (curved outwards, like the back of a spoon). When you look at your face in the inside of a shiny spoon, the image might appear inverted, while on the outside it looks smaller and upright.

Concave mirrors reflect light inwards and can form images that are enlarged, diminished, inverted, or erect depending on the object's distance from the mirror. In contrast, convex mirrors always form smaller, erect images and give a wider view. That's why cars use convex mirrors as side-view mirrors—to show more area.

Observing Images with Spherical Mirrors

When an object is close to a concave mirror, its image appears large and upright. Move the object farther, and the image becomes smaller and upside down. With convex mirrors, no matter how far or close the object is, the image stays upright but is always smaller than the actual object. Plane mirrors, unlike spherical ones, produce erect images of the same size as the object.

Here are some practical uses of spherical mirrors:

- **Concave Mirrors**: Used in dentist's tools to enlarge view, torches and headlights to focus light, and telescopes for collecting light from stars.
- **Convex Mirrors**: Used for surveillance in stores, at road intersections for a wider view, and as vehicle side mirrors for safety.

Laws of Reflection

Reflection is when light bounces off a surface like a mirror. There are two main laws of reflection:

- The angle of incidence (angle between incoming ray and a line perpendicular to the mirror) is equal to angle of reflection (angle between outgoing ray and the normal).
- The incident ray, the normal, and the reflected ray all lie in the same plane.

These rules apply to all kinds of mirrors, whether plane, concave, or convex.

If several parallel light beams fall on a mirror:

- **Plane mirror:** Beams reflect in parallel directions.
- **Concave mirror:** Beams converge (come together at a point).
- **Convex mirror:** Beams diverge (spread apart).

This makes concave mirrors useful for concentrating light—like in solar concentrators and solar cookers.

What are Lenses? Types and Image Formation

Lenses are transparent objects with curved surfaces, usually made of glass or plastic. There are two main types:

- **Convex lens:** Thicker in the middle than at the edges—also called a converging lens because it brings light rays together. A drop of water also acts as a convex lens.

- **Concave lens:** Thinner in the middle and thicker at the edges—called a diverging lens because it spreads light rays apart.

Behavior and Uses of Lenses

Looking through a convex lens at a nearby object, you see an enlarged and upright image. As you move the object farther from the lens, the image turns upside down and gets smaller. In contrast, images seen through a concave lens are always upright and smaller than the actual object, regardless of distance.

Convex lenses converge parallel light beams to a single point—helpful for focusing sunlight to ignite paper (with safety precautions). Concave lenses spread out beams. These properties make lenses essential in:

- Eyeglasses (correcting vision)
- Cameras and smartphones (focusing images sharply)
- Telescopes and microscopes (magnifying distant or tiny objects)
- Solar devices for generating heat

Even the human eye uses a convex lens to focus light on the retina.

Key Points and Summary Table

- **Concave mirrors:** Can produce enlarged, diminished, inverted, or erect images depending on distance.

- **Convex mirrors:** Always produce small, upright images.

- **Convex lenses:** Image may be large and upright (nearby object) or small and inverted (distant object).
- **Concave lenses:** Always produce small, upright images.
- Laws of reflection work for all mirrors.

Here is a quick comparison:

Type	Image Nature	Usual Uses
Concave Mirror	Large & Upright (close), Inverted (far)	Dentists, torches, telescopes
Convex Mirror	Small, Upright (always)	Vehicle mirrors, surveillance
Convex Lens	Large & Upright (close), Inverted (far)	Magnifiers, cameras, eyes
Concave Lens	Small, Upright (always)	Eyeglasses (for near-sightedness)

Practice and Explorations
To strengthen understanding, try class activities like observing your image in spoons, drawing rays on paper, or using lenses and light beams. Think about why convex mirrors are safer for drivers and how telescopes use concave mirrors. Explore matching exercises or try using a magnifying glass with text for hands-on learning.

By practicing and revising these concepts, you will be well prepared for your exams and for understanding the world of optics in daily life!